

# Chapter 5 Data compression

ch 5-1 - 5.6

Information → Code  
Data 4 symb

Morse code: Alphabet → seq of symb

$e \rightarrow C(e) = \cdot$

$a \rightarrow C(a) = -$

most freq letter → shortest

least " → longest

e i t a n ... p r z u y  
short code long code

Object: given  $X$ : words w  
P.M. f  $P(x)$

## §5.1 Examples of codes

$D^*$ : sequence of  $D$ -ary alphabet

$D = \{0, \dots, D-1\}$

2-ary  $D = \{0, 1\}$

Def Source code  $C$

$C: X \rightarrow D^*$

$\downarrow$   
 $x$

$\downarrow$   
 $C(x)$

$C(x) = 0$

$\downarrow$   
 $00$   
 $\downarrow$   
 $11$

$l(x)$ : length of  $C(x)$

Ex  $X = \{\text{red, blue}\}$

$C(\text{red}) = 00$

$C(\text{blue}) = 11$

$l(\text{red}) = 2$

$l(\text{blue}) = 2$

Def Expected length  $L(C)$  of  $C$ .

for random  $X$ . p.m.f  $P(x)$ ,

$$L(C) = \sum_{x \in X} P(x) l(x)$$

Ex 5.1.1  $X$ : rand. var with

|   | $P_r(X=x)$    | $C(x)$ | $l(x)$ |
|---|---------------|--------|--------|
| 1 | $\frac{1}{2}$ | 0      | 1      |
| 2 | $\frac{1}{4}$ | 10     | 2      |
| 3 | $\frac{1}{8}$ | 110    | 3      |
| 4 | $\frac{1}{8}$ | 111    | 3      |

|               |                                                       |
|---------------|-------------------------------------------------------|
| $H(X) = 1.95$ | 0 11 0 11 11 00 11 0                                  |
| $L(C) = 1.75$ | <u>1</u> <u>3</u> <u>4</u> <u>2</u> <u>1</u> <u>3</u> |

$$1. H(X) = \underline{L(C)}$$

2. Uniquely decodable,

Ex 5.1.2

| $x$ | $P_r$         | $C(x)$ | $l(x)$ |
|-----|---------------|--------|--------|
| 1   | $\frac{1}{3}$ | 0      | 1      |
| 2   | $\frac{1}{3}$ | 10     | 2      |
| 3   | $\frac{1}{3}$ | 11     | 2      |

5-2

$$H(X) = 1.58$$

$$H(X) \leq \underline{L(C)}$$

$$L(C) = 1.666$$

$$1. H(X) < L(C)$$

2. Unique decodable

Def A code is nonsingular

if  $\forall x, x'$

$$x \neq x' \Rightarrow C(x) \neq C(x')$$

(1-1 mapping)

$$C(x) = 00$$

$$C(x') = 01$$

$$x_1 x_2 \dots x_n \rightarrow C(x_1) C(x_2) \dots C(x_n)$$

Def The extension  $C^*$  of a code  $C$

$$C(x_1 \dots x_n) = C(x_1) \dots C(x_n)$$

$$\text{Ex } C(x_1) = 00$$

$$C(x_2) = 11$$

$$\Rightarrow C(x_1 x_2) = 0011$$

Def A code is Uniquely decodable

if its extension is nonsingular

$x_1 \dots x_n : 1-1 \text{ mapping}$

$(11010 \dots 10 \dots 1000) \rightarrow$   
 $\rightarrow$

Instantaneously vs read all

Def A code is a prefix code

or an instantaneous code

if no code word is a prefix of any code word.

Ex

$$C(x_1) = 11$$

$$C(x_2) = 110$$

not prefix

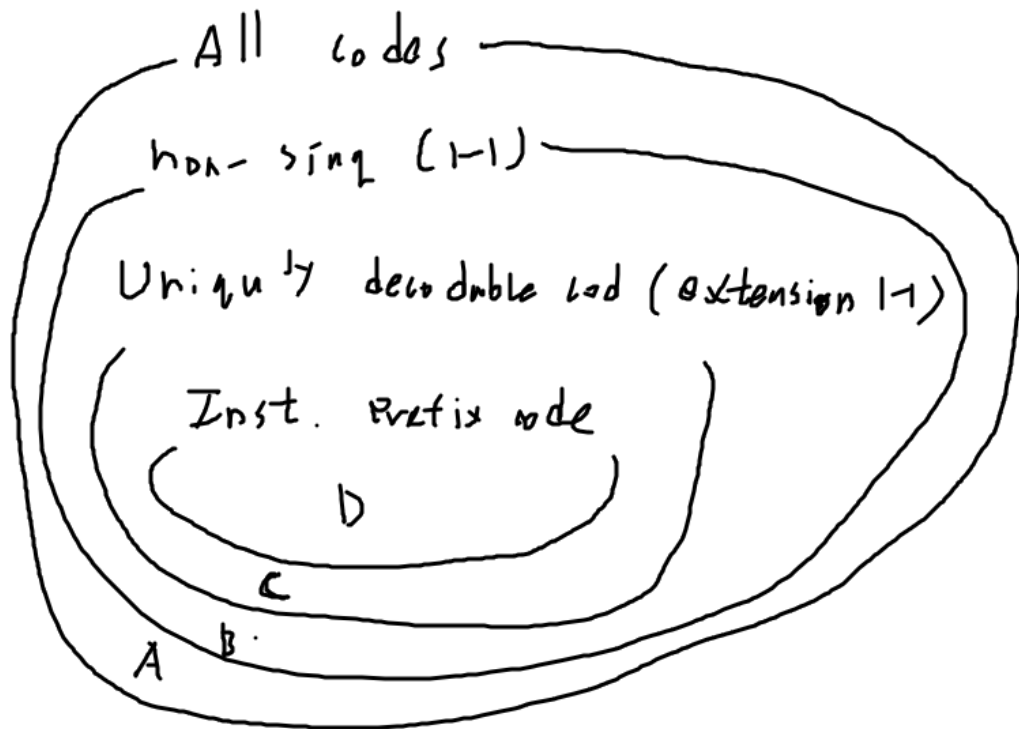


Table 5-1

|   | A            | B   | C. ↔ D                           |
|---|--------------|-----|----------------------------------|
| X | non singular |     |                                  |
| 1 | 0            | 0.  | 10 0                             |
| 2 | 0            | 010 | 00 10                            |
| 3 | 0            | 01  | 11 11                            |
| 4 | 0            | 10  | <u>110</u> <sup>prefix</sup> 111 |

1-1. (1100...)

010  
010 → 2  
010 → 14  
 31

3 2 ...  
 1100  
4 2

1100...  
3 2

1 What is the most efficient (shortest)

int. code → § 5.2 → 5.4

2 How can we construct inst. code

§ 5.6.

# { 5.2 Kraft Ineq

(lower bound of int. code)

Thm 5.2.1. For inst. code over an alphabet size  $D$ .

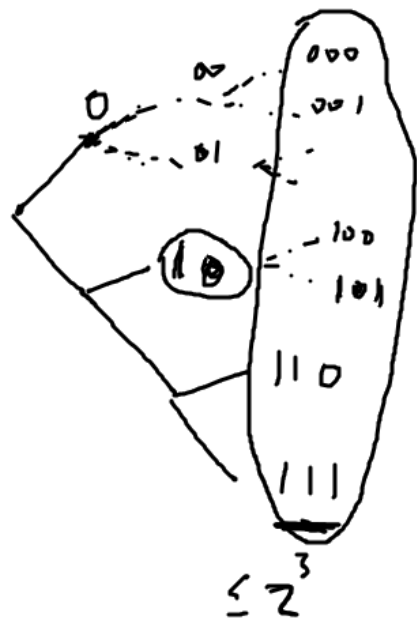
code word length  $l_1, l_2, \dots, l_m$  must satisfy  $\frac{1}{D^{l_1}}, \frac{1}{D^{l_2}}, \dots, \frac{1}{D^{l_m}}$

**Kraft**  $\sum_{i=1}^m D^{-l_i} \leq 1 \quad D^{-l_1} + \dots + D^{-l_m} \leq 1$

Converse. If a set of code word length satisfy Kraft Ineq

$\Rightarrow \exists$  inst code.

Pf) 2-ary tree Fig 5.2



# eliminate

|                   |           |
|-------------------|-----------|
| 0's desc          | $2^{3-1}$ |
| 10's desc         | $2^{2-1}$ |
| 110 $\rightarrow$ | $2^{3-3}$ |
| 111 $\rightarrow$ | $2^{3-3}$ |
| +                 |           |

$\leq 2^3$

$\Rightarrow \sum_{i=1}^m 2^{l_{max} - l_i} \leq 2^{l_{max}}$

$\Rightarrow \sum_{i=1}^m D^{l_{max} - l_i} \leq D^{l_{max}}$

$\Rightarrow \sum_{i=1}^m D^{-l_i} \leq 1$

Thm 5.2.2 (Extended Kraft inequality)

For any countably infinite set of codewords of inst. code

$$\sum_{i=1}^{\infty} D^{-l_i} \leq 1$$

Converse: true.

Pf) we 2-ary alphabet,  $y_i$ : code word.

$$\text{Let } 0.y_1 \dots y_{l_i} = \sum_{j=1}^{l_i} y_j \cdot \underbrace{D^{-l_i}}_{\substack{-0.1 \\ D^{-l_i}}}$$

we define

$$[0.y_1 \dots y_{l_i}, 0.y_1 \dots y_{l_i} + \frac{1}{D^{l_i}})$$

For example

$$C(y_1) = \frac{10}{2}$$

$$0.10 = 10 \cdot \frac{1}{2^2} = 10.0.0) \\ = 0.10$$

$$[0.10, 0.10 + \underbrace{\frac{1}{2^2}}_{\substack{0.01 \\ 0.01}})$$

$$[0.10, 0.11) \quad \begin{array}{cc} 101 & 1001 \\ 100 & 1000 \end{array}$$

: includes all code word begin with 10

$$\cup [ ) \subset [0, 1)$$

$$\sum_{i=1}^{\infty} D^{-l_i} \leq 1$$

□

# { 5.3 Optimal code

Object : Finding inst. code  
with minimum  $E_L(x)$

$l_i \in \mathbb{R}$ .

Minimize  $L = \sum p_i l_i$   
satisfying  $\sum D^{-l_i} = 1$

Lagrange multiplier

$$J = \sum p_i l_i + \lambda \left( \sum D^{-l_i} \right)$$

$$\Rightarrow \frac{\partial J}{\partial l_i} = p_i - \lambda D^{-l_i} \log_e D = 0$$

$$D^{-l_i} = \frac{p_i}{\lambda \log_e D}$$

$$1 = \sum D^{-l_i} = \sum \frac{p_i}{\lambda \log_e D} = \frac{1}{\lambda \log_e D}$$

$$\lambda = \frac{1}{\log_e D}$$

Optimal length  $\Rightarrow \underline{l_i^*} = -\log_D p_i$

Expected length  $\Rightarrow L^* = \sum p_i l_i^*$

$$= -\sum p_i \log_D p_i$$

$$:= H_D(x)$$

Thm 5.3.1

The expected length  $L$  of inst

$D$ -ary code satisfy

$$L \geq H_D(x)$$

Equality holds when  $D^{-l_i} = p_i$

$$\text{Pf)} \quad L - H_D(x) = \sum p_i \underline{l_i} - \sum p_i \log_D p_i$$

$$= - \sum p_i \log_D D^{-l_i} - \sum p_i \log_D p_i$$

$$= \sum p_i \log_D \left( \frac{D^{-l_i}}{p_i} \right)$$

we define  $v_i = \frac{D^{-l_i}}{\sum_j D^{-l_j}} = 1$

$$D(P||v) \geq 0$$

$$L - H_D(x) = \sum p_i \log_D \left( \frac{p_i}{v_i} \right) \leq 1$$

$$- \sum p_i \log_D \left( \sum D^{-l_i} \right) \geq 0$$

$$\geq 0$$

Equality  $D(P||v) = 0 \quad \frac{p_i}{v_i} = 1$

$$\sum D^{-l_i} = 1$$

$$\frac{p_i}{v_i} = \frac{p_i}{D^{-l_i}} = 1$$

$$p_i = D^{-l_i}$$

$$p_i = D^{-l_i}$$

$$\Leftrightarrow l_i = -\log_D p_i \quad \square$$



Def A probability dis is called  
D-adic if each prob is  
equal to  $D^{-n}$  for some  $n$ .

X: D-adic  $\Rightarrow -\log_D p_i = \text{integer}$

$$L_i^* = n.$$

§ 5.4. Suboptimal procedure (Shannon-Fano)

§ 5.6. optimal " (Huffman)

§ 5.4. Bounds on the  
optimal code length

$$L_i = -\log_D p_i \neq \text{integer}$$

$$L_i \equiv \left\lceil \log_D \frac{1}{p_i} \right\rceil$$

Tr: smallest integer  $x \geq 0$

$$\log_D \frac{1}{p_i} \leq \left\lceil \log_D \frac{1}{p_i} \right\rceil$$

$$\Rightarrow \sum_i D^{-\left\lceil \log_D \frac{1}{p_i} \right\rceil} \leq \sum_i D^{-\log_D \frac{1}{p_i}} = 1$$

$$-\log_2 \frac{1}{p_i} \leq \underline{\underline{l_i = \lceil \log_2 \frac{1}{p_i} \rceil}} < \log_2 \frac{1}{p_i} + 1$$

$$\sum_i p_i$$

1 bit.

$$\Rightarrow \underline{\underline{H_D(X) \leq L < H_D(X) + 1}}$$

$$\underline{l_1^* - l^*}$$

Thm 5.4.1 Let  $l_1^*, \dots, l_m^*$  optimal code lengths.

Then  $H_D(X) \leq L^* < H_D(X) + 1$

Pf)  $l_i^* \leq l_i$

$$L^* \leq L$$

$$\underline{H_D(X) \leq L^*} \leq L < H_D(X) + 1$$

Thm 5.3.1

(1)

Let  $X$ : rand var  $p_1, \dots, p_n$

$p_1, \dots, p_n$

$\Rightarrow$  error.  $\Rightarrow D(P||Q)$

Thm 5.4.3. The expected length under  $P(x)$  of the code

$$\text{length } l(x) = \left\lceil \log_2 \frac{1}{p(x)} \right\rceil$$

$$H(P) + \underline{D(P||Q)} \leq E_P l(x)$$

$$\textcircled{\leq} H(P) + \underline{D(P||Q)} + 1$$

$$\text{pf "}\leq\text{"}. E l(x) = \sum p(x) \left\lceil \log_2 \frac{1}{p(x)} \right\rceil \leq \sum p(x) \left( \log_2 \frac{1}{p(x)} + 1 \right)$$

$$= D(P||q) + H(P) + 1$$

$$\text{"Z"} \quad E_L(x) > \sum_x p(x) \log \frac{1}{q(x)}$$

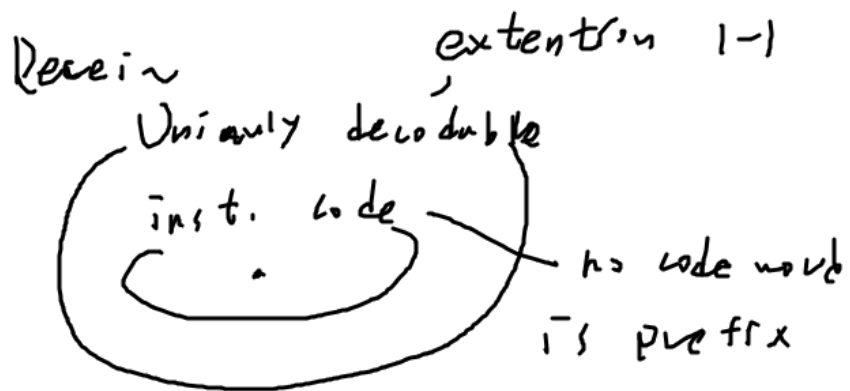
$$= \sum_x p(x) \log \frac{p(x)}{q(x)} \frac{1}{p(x)}$$

$$= D(P||q) + H(P)$$

where  $H(P)$  gives penalty  $H(P)$

$$\underline{D(P||q)}$$

§ 5.5 Kraft inequality for Uniquely decodable codes



§ 5.2 inst code  $\Leftrightarrow$  Kraft  $\sum D^{-l_i} \leq 1$

§ 5.5 Uni. decodable  $\Leftrightarrow$  1

Unique decodable  $\stackrel{\text{lengths}}{=} \text{inst.}$

Thm 5.5.1 The code word lengths of Uniquely decodable D-ary code must satisfy

$$\sum D^{-l_i} \leq 1$$

Pf) For extension code

$$l(x_1 \dots x_n) = \sum_{i=1}^n l(x_i)$$

$$\left( \sum_{x \in X} D^{-l(x)} \right)^K = \sum_{x_1 \in X} \dots \sum_{x_n \in X} D^{-\overbrace{l(x_1) + \dots + l(x_n)}^K}$$

$1, 2, \dots, K$  lengths

$$= \sum_{\substack{m=1 \\ \leq D^m}}^{K \cdot \text{lengths}} a(m) D^{-m}$$

$a(m)$  : # of code word with length  $m$   
 $a(m) \leq D^m$

$$\left( \sum_{i \in N} D^{-l_i} \right)^k \leq k l_{\max}^{\frac{1}{k}}$$

$$\sum_{i \in N} D^{-l_i} \leq \underbrace{(k l_{\max})^{\frac{1}{k}}}$$

$$\rightarrow 1$$

$$\text{as } k \rightarrow \infty$$

$$\sum D^{-l_i} \leq 1 \quad \underline{\square}$$

Conclusion.

$$\text{Union de } \infty \text{ d'able } \stackrel{\text{length 1}}{=} \text{Inst}$$

# § 5.6 Huffman code

Inst.

Ex 5.6.1 For 2-ary alphabet

(a)

| X | $P_r(X=x)$ |     |             |      |             |    |          |
|---|------------|-----|-------------|------|-------------|----|----------|
| 1 | 0.25       | 11  | 0.3         | 10   | <u>0.45</u> | 0  | 0.45 - 0 |
| 2 | 0.25       | 00  | 0.25        | 11   | 0.3         | 10 | 0.55 - 1 |
| 3 | 0.2        | 01  | <u>0.25</u> | 00   | 0.25        | 11 |          |
| 4 | 0.15       | 0.3 | 0.2         | 0.45 | 01          |    |          |
| 5 | 0.15       | 100 |             |      |             |    |          |
|   |            | 111 |             |      |             |    |          |

Ex 5.6.3 not 2-ary D 23  
 D: 3-ary alphabet

$$1 + k(D-1)$$

$$1 + k \cdot 2$$

3-ary  
 $\nearrow$

X  $P_r(X=x)$

1 0.15 1 0.25 1 0.5 - 0

2 0.25 2 0.25 2 0.25 - 1

3 0.2 01 0.2 00 0.25 - 2

4 0.1 02 0.2 01

5 0.1 000 0.1 02

6 0.1 001

dummy

0

002

17